

WEST TEXAS TOWN MEETING

On Electricity Issues

The economy of West Texas has been steadily improving over the last several years. Consequently, the need for electricity also has been steadily increasing. The last power plant added to the West Texas Utilities Company (WTU) electric system was in 1987. Given the increase in demand for electricity in the retail and wholesale markets, WTU will need additional resources (approximately 150 - 250 megawatts of electricity in the 1998 - 2001 time period) to meet the increasing demand for electricity. There are several ways to meet this need, and no one simple or obvious answer. So WTU, in consultation with you, its customers, must decide the best way to respond to this anticipated need.

About the Town Meeting

Your participation in the West Texas Town Meeting will give WTU the opportunity to learn what customers think about the issues that arise from meeting future energy needs. This pamphlet is designed to introduce these issues:

- WTU's projected energy needs
- The choices in meeting those needs
- The costs associated with each choice
- Other considerations

As participants in the town meeting, you will have an opportunity to discuss the issues yourselves in both large and small group settings. You also will have an opportunity to ask questions of the different groups who have an interest in the outcome of the process. After reflecting on the different perspectives and options, you will develop your own judgments based upon those things that you find important. The views of the participants are very important because they will represent the views of all 187,000 WTU customers.

Saturday afternoon will include a discussion with representatives from the Public Utility Commission of Texas. This session will be taped for television broadcast by KTXS News 12 at a later date.

WTU's Planning Process

Recent changes in Texas law require a utility that has a need for new power plant capacity to compare all the available options in an integrated resource plan, commonly called an IRP. The law also directs utilities to develop a portfolio of resources, meaning the best mixture of resources to meet needs at an affordable price.

A part of the IRP process is involving the public to gather information on local concerns about the environment, the effects of utility operations on the local culture and economy and what customers expect from their utility in terms of cost and services. This public participation is to take place at the beginning of the IRP process so the results can be incorporated throughout the regulatory process.

What are the choices?

There are four basic ways to meet future increases in energy needs:

- Increase Efficiency
- Buy Power and Transport
- Build a Fossil Fuel Plant
- Build Renewable Energy Plants

Some particular combination may prove to be the best approach. There will be several questions on the table as you deliberate. What do you think about the costs and benefits of various options? What are your most important concerns in choosing among the options? How much do you think customers should pay—now and in the long run—for their increasing electricity needs?

As a participant in the West Texas Town Meeting on Electricity Issues, you will play a crucial role in WTU's plans.

Costs and Benefits of Options

Deciding the best course of action is not going to be easy. Every reasonable option has both advantages and disadvantages. What one person considers a good solution, another may not. To get more of a trait you like, you may have to accept more of something you don't like.

Fair evaluation of the alternatives means considering not only the dollar cost of each option but other factors as well. These factors include reliability, potential impacts on the environment, economy and society, as well as the availability of the fuel.

Increase Efficiency

One option to meet electricity needs is to use the power that we already have more efficiently. While it would be very challenging for this

WTU's
West Texas
Town Meeting
on Electricity Issues
August 9 and 10
Hardin-Simmons
University
at
Abilene

approach to meet WTU's entire need for new resources, it can play an important role.

Some energy efficiency programs are designed to reduce overall energy use. Others focus specifically

on getting customers to use less electricity during peak usage hours when the greater need for electricity drives costs higher and causes the need for more electrical facilities.

Buy Power and Transport

Another way to meet the need is to buy electricity from other sources instead of WTU generating it. Surplus power is available from industrial companies that generate more than they need and from other electric utilities who have excess electricity.

Currently there is a surplus of power in Texas and this surplus is expected to continue for the next few years. This makes the wholesale power market in Texas very competitive in the 3-5-year time frame, but the long-term market is less certain.

Build a Fossil Fuel Plant

Another option to meet resource needs is to increase the region's electricity production. This would entail WTU or some other company building some type of new power plant. A new power plant could be a traditional fossil fuel power plant using natural gas or coal, a smaller natural gas or diesel plant, or small plants using a new technology. These new emerging technologies could include options such as fuel cells, energy storage devices and distributed generation.

Build Renewable Energy Plants

A fourth option is to build plants that use renewable energy sources. Renewable energy could play an important role in meeting WTU's energy needs. WTU could tap those sources of energy that are readily available: wind and solar energy. Renewable energy provides power without consuming fossil fuel, thus, also benefiting the environment and protecting customers from possible swings in fuel price and availability.

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WTU and its Stakeholders

Groups affected by energy decisions are referred to as stakeholders because they each have a stake in the outcome. Each stakeholder group may benefit from a different solution. Our challenge is to find the best set of options that considers all customers' viewpoints and is the greatest value.

WTU is a regulated monopoly. That means that, in exchange for the right to provide electricity within its service territory, WTU has an obligation to serve all customers who wish to purchase the company's service. WTU's status as a regulated monopoly also means that the company must follow the rules and procedures of the Public Utility Commission of Texas. WTU must also respond to a number of other groups that believe that the company's operations have an impact on issues about which they are concerned.

WTU's Service Territory

Based in Abilene, Texas, West Texas Utilities Company (WTU) supplies electric service to more than 187,000 customers in a 53,000-square-mile area. WTU serves 170 towns and communities, including Abilene and San Angelo, throughout a 52-county area of West Texas. WTU's service territory is primarily rural with agriculture and oil production serving as the main industries.

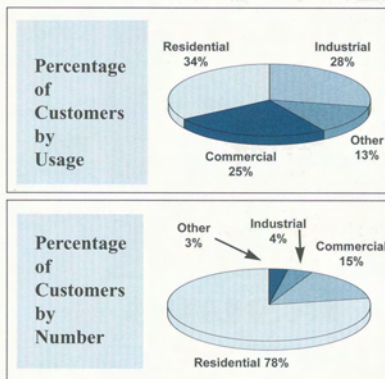
WTU currently has approximately 1,426 megawatts of generating capacity and more than 6,500 miles of transmission lines to serve its customers' needs. A megawatt of generating capacity is enough to serve approximately 250 average homes.

WTU's Customers

WTU has three main groups of retail customers: residential, commercial and industrial. Participants in the town meeting reflect a random sample of each of these groups. WTU also serves a number of customers at the wholesale level, such as municipal electric systems and electric cooperatives.

Residential Customers - WTU had over 143,000 residential customers at the end of 1995. The average residential customer uses 850 kilowatt-hours of electricity and pays approximately \$60.00 a month when averaged out over the entire year. WTU's total number of residential customers can increase or decrease during the year.

Commercial Customers - This group includes businesses such as supermarkets and other types of stores, restaurants, office build-



ings, schools and churches. The amount of electricity that these groups use can differ greatly depending on the size of the facility, the type of heating and cooling equipment they use and any special equipment such as freezers, cooking equipment and mainframe computers.

Industrial Customers - This group includes customers such as oil pumping and refineries. These customers use large amounts of electricity, and it is not uncommon for these customers to operate 24 hours a day.

The town meeting will include a representative sample of each type of retail customer. Each group of customers has particular needs and patterns of using electricity. Therefore, different types of customers might lean toward different options. For example, residential customers tend to use more electricity from 4 p.m. to 6 p.m. each day. Usage by industrial cus-

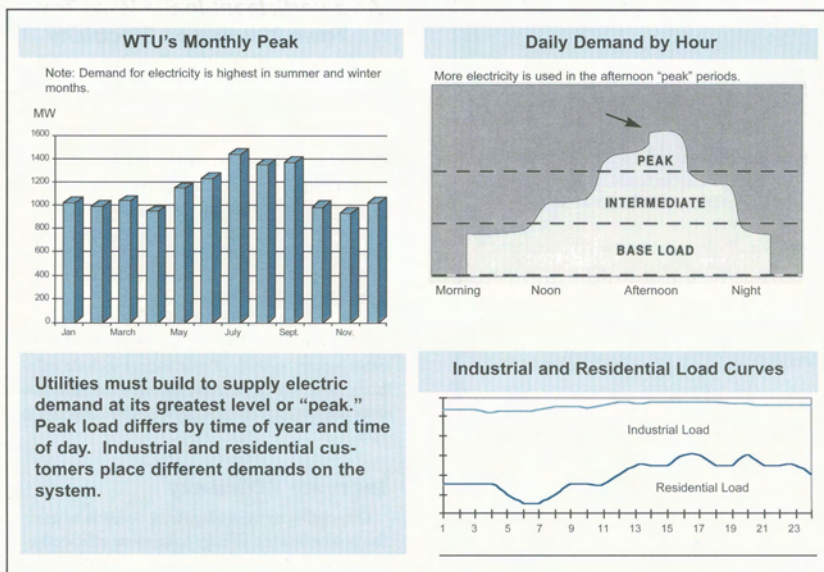
tomers tends to be more constant without the peaks. (See the charts below.)

Wholesale Customers - WTU has traditionally served both the retail market and the wholesale market in West Texas. The wholesale electricity market is made up of cities and rural electric coops that purchase power from WTU and then resell it to their own retail customers. As much as 30 percent of the power WTU produces goes into the wholesale market, which is now very competitive. Wholesale customers can now choose their own electric supplier; and although WTU has served these customers in the past, it is unknown how much electricity WTU will sell to them in the future.

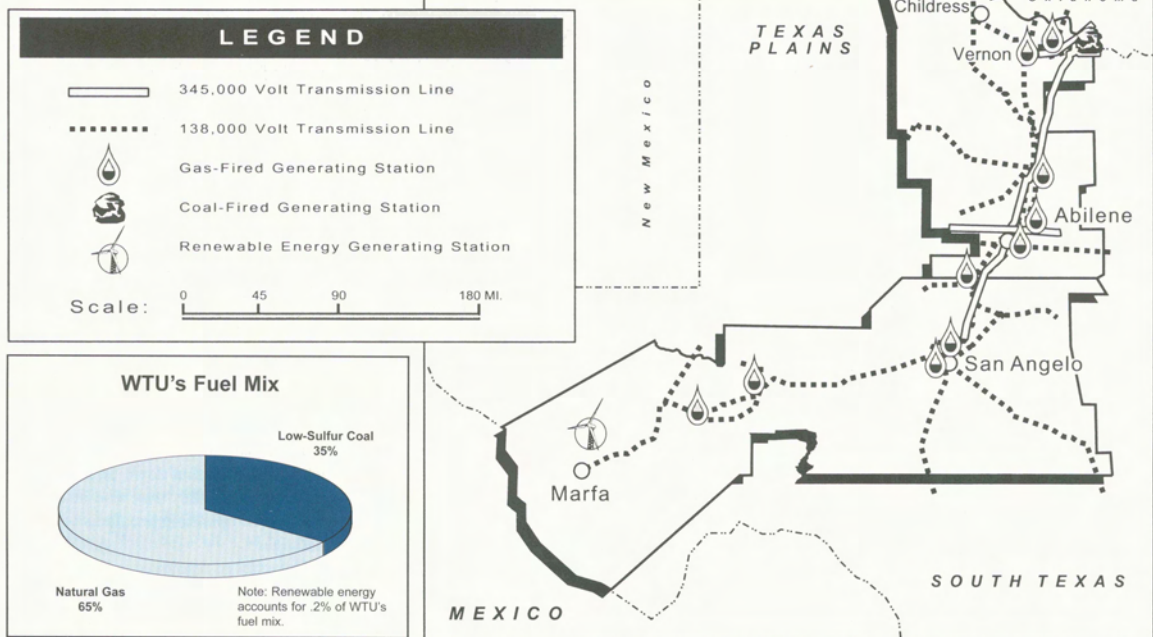
WTU and Its Shareholders

WTU is a wholly owned subsidiary of Central and South West Corporation (CSW), an electric utility holding company with corporate headquarters in Dallas. There are approximately 201 million shares of CSW common stock outstanding including shares owned by WTU customers. In addition to WTU, CSW holds all of the common stock of Public Service Company of Oklahoma, Southwestern Electric Power Company and Central Power and Light Company. These companies provide electric service to 1.6 million customers in a service area covering 152,000 square miles, the second largest area served by an electric utility system in the United States. CSW also owns SEEBORAD plc, a regional electric distribution company that serves 2 million customers in England.

Electric Load Varies Over Time



WEST TEXAS UTILITIES COMPANY SERVICE TERRITORY



Competition

There are others capable of providing electric service within WTU's service territory. Currently, these groups include other electric utilities, electric cooperatives, independent power producers and self-generators. Propane suppliers could also be considered competitors. Changes in federal and state laws may allow even more competition in the future among providers of electric service. Competitors are interested in how WTU plans to meet its future energy needs. These groups will have the opportunity to supply WTU's additional electricity needs when WTU issues a Request for Proposals (RFP) to meet the need.

There is currently a movement by some groups to deregulate the electric industry similarly to deregulation that has occurred in the telephone and natural gas industries. The debate on competition in the electric utility industry is ongoing and occurring in many different forums. While we can't predict the final outcome, regulators at the state and federal levels are moving to encourage competitive markets.

Governmental Agencies

PUCT

The Public Utility Commission of Texas (PUCT) regulates WTU and about 160 electric and telephone utilities throughout the state, including eight large investor-owned electric utilities like WTU.

The PUCT's four basic responsibilities are:

setting reasonable rates, approving construction of new power plants and transmission lines, monitoring safety and reliability of utility service and resolving customer complaints.

Office of Public Utility Counsel

The Office of Public Utility Counsel (OPUC) was established in 1983 to provide representation for residential and small business customers. OPUC participates in most major rate cases, rule-making proceedings and certification cases before the PUCT.

FERC

The Federal Energy Regulatory Commission (FERC) regulates all interstate wholesale electric power. The FERC is currently at the forefront of the move to promote wholesale electric utility competition. FERC rules now require utilities that own transmission lines to transport power for other utilities at the same rate they charge themselves.

Other Groups

There are a number of other groups that take an interest in how WTU plans to meet its future energy resource needs. Several of these groups are listed below:

Environmental Groups

Environmental Defense Fund
Sierra Club
Audubon Society
Texas Citizen Action
Sustainable Energy Economic Development

Consumer Groups

Texas Legal Services
Public Citizen
Consumers Union
Local Taxpayers Associations
Texas Ratepayers' Organization
to Save Energy (Texas Rose)
Texas Industrial Energy Consumers (TIEC)

Other Parties

Cities
State Agencies
Wholesale customers

Typical Residential End Uses of Electricity

End Use	Percent of Total
Heating	3.5%
Cooling	21.0%
Ventilation	2.4%
Dishwasher	1.4%
Clothes Washer	0.5%
Water Heater	13.8%
Cooking	3.5%
Refrigerator	10.9%
Freezer	4.3%
Dryer	7.1%
Misc.	31.2%
Total	100.0%

Issues for Consideration

Different groups represent different priorities; many of which you may share. Issues primarily break down into costs and long-term social goals. Central to the debate is whether resources should be selected based on today's cost or what we think their cost will be in the future. Unpredictable fuel prices, rising environmental standards, health and other public concerns, such as the local and state economy, are all part of the equation.

Dollar Cost of Resources

The cost of electricity is determined by a combination of factors. Some of these include capital equipment cost, operation and maintenance expense, and fuel expense. Fuel is a major cost of the production of electricity today. The fuel expense can vary from 25 percent to 50 percent of customer bills.

Some customers, especially those facing stiff competition in their businesses, or those on fixed incomes believe the most important factor in selecting a resource should be cost. They say that while other priorities might be "nice to have," they prefer not to pay the added costs. In the short-term versus long-term cost debate they argue they must first survive today in order to worry about tomorrow. Their preference is that new generation resources should be put out for bid and the lowest bid should be selected.

Some of these customers argue for more competition in the supply of electricity and are willing to sort through the competing offers for the best deal, whether it is from their existing supplier or some other source.

Another group of customers favor lowest cost, but are interested in looking at both short-term and long-term costs. One of their major concerns is the unpredictability of future fuel costs. They are willing to look at ways to hedge the risk of rising natural gas prices.

Environmental Costs

Any electrical generation process that burns a fuel creates emissions. Some of the emissions are controlled by national standards and others are not. Any plant built must meet all current state and federal environmental standards. However, this does not eliminate all pollution. The company can make additional pollution control investments. One unknown is that standards may increase in the future if environmental laws change or environmental problems increase. The major pollutants are as follows:

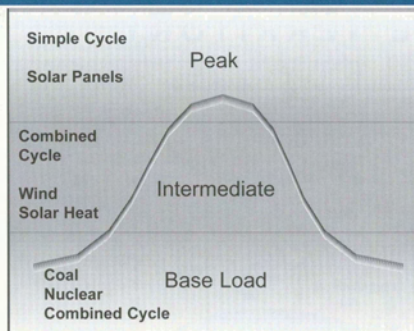
Oxides of Nitrogen - Component of smog and ozone

Sulfur Dioxide - Leads to acid rain

Carbon Dioxide - Greenhouse gas that can lead to climate change. CO₂ is not currently controlled.

Fine Particles - Lead to haze and can affect breathing.

Many environmental and consumer groups



Generation Plants can be separated into three major categories: Base Load, Intermediate and Peak. Base load plants typically are larger power plants that run most of the time. Intermediate plants run 30 to 60 percent of the time. Peak Plants operate during those times when electricity demand is at its highest point. At left is a usage curve that shows the relationship to the three categories along with the types of plants that normally fall into those categories.

state that power plants in Texas, even those in full compliance with environmental laws are a major source of air emissions. According to the Environmental Protection Agency, U.S. power plants contribute about 72 percent of sulfur dioxide emissions, 33 percent of nitrogen oxide emissions, 32 percent of particulate emission, and 36 percent of man-made carbon dioxide emissions.

Some advocates of the IRP planning process believe that the utility should take possible cleanup costs and health costs to residents above those required by law into account when selecting resources. In addition, they are concerned about rising costs of environmental controls. Finally, environmental advocates are concerned about impacts on climate due to greenhouse gas emissions.

Economic Impact

Any choice to add resources to the process will result in some effect on employment and business growth. Should WTU take these kinds of effects into account in its evaluation of resources? Some believe that building fossil fuel plants would be a boost to the area's economy through additional jobs and increasing the tax base. Others believe that stimulation of the energy efficiency and renewable energy industries is best for the economy. In the long term, they say these resources will help to make Texas more energy independent, add new industries and create more new jobs for Texans than a fossil fuel plant.

Others believe that the least expensive electric service will attract business, thereby boosting the economy. Some customers do not believe utilities should be using customer money to engage in economic or social policy.

Reliability

WTU and other utilities are required by law to provide their customers with a firm supply of electricity. This means that electricity must be available when customers want to use it. There is one exception. A customer may voluntarily accept a lower rate in exchange for interruption of its power supplies. Under these agreements, customers receive rate discounts.

A high level of reliability is maintained through several measures. National reliability councils set standards for lowering the risks of outages. Utilities maintain a reserve margin of

capacity (15 percent more than they need) to meet high peak demands. Power can be transported between service territories in an emergency.

Reliability standards are necessary because no source of power is available all the time. Even though they run most of the time, base load plants (coal, nuclear and natural gas combined cycle) must be shut down on a schedule for maintenance. They may also be out of service because of operating problems or storm damage.

Diversity of resources is critical to the reliability of the system. The idea is to "put your eggs in more than one basket" to minimize risk. Smaller plants are better for reliability than large plants. If a small plant fails, the shortfall can easily be covered by intermittent or peaking resources that are not currently serving load.

As a general rule, smaller sources of energy (or energy savings) improve system reliability. This is because a single breakdown from a small source presents no threat to providing continuous power for customers.

Fuel and Resource Mix

The availability of fuel is critical to maintaining system reliability. Once again, diversity — not having all your eggs in one basket — is beneficial. This diversity can be achieved by balancing the fuels used for generation. Fuel and power purchases can be short-term (priced at the "going rate") or long-term (secured with long-term, average price contracts). In the case of solar and wind energy, availability is dependent on natural forces, such as weather, as well as careful placement of generating plants.

WTU's fuel mix is made up of firm gas, spot gas and long-term coal contracts. The coal and natural gas industries have predicted that supplies will be adequate for the foreseeable future and are willing to sign long-term contracts in exchange for more stable prices than those in the spot market. Historically, the market place for these fuel supplies has seen spikes in prices and, at times, disruptions in fuel supply.

Consumer and environmental groups believe that the best way to hedge bets on fuel availability and fuel price is to diversify the energy mix with energy efficiency and renewable resources. Energy efficiency and renewable resources are fuel-free, and therefore not subject to market price swings.

Regulatory Rate Setting

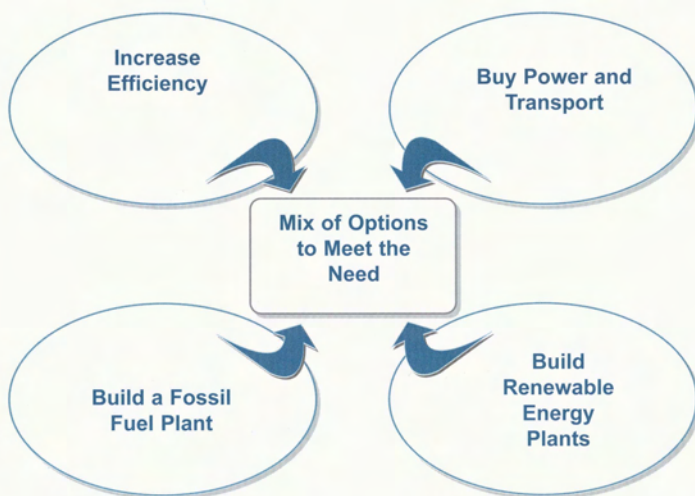
How the Utility Earns a Profit

State laws require that electric utilities serve every customer who requests electric service. In return, the electric utility is granted an opportunity to earn a reasonable profit on the money invested to serve those customers and to recover its reasonable operating expense. In setting rates, the Public Utility Commission of Texas (PUCT) approves an annual revenue requirement that is based on the costs incurred by WTU in the previous 12 months. The costs are divided among customer groups (residential, commercial and industrial) and split into rates that are used to figure customer bills. A mild summer can cause a revenue deficit. A hotter than normal summer can cause a revenue gain.

Predictable profits are earned only on the dollars a utility invests that are determined to be reasonable by the PUCT. These include funds used to build power plants and transmission lines and some energy efficiency program expenses. Fuel expenses and purchased power costs are collected from customers without a markup when approved by the PUCT as being reasonable.

The utility business is changing as competition increases. Currently, the utility selects the lowest cost option whether or not they are allowed to earn a profit on the transaction. Many utilities believe incentives would allow them an opportunity to earn a profit for their shareholders on purchased power transactions when the power comes from another company's plant. They argue they provide value as a broker and incur some risk. Consumer advocates believe that utility companies are required by law to diversify their resources and should not be paid to do so. They also have concerns that automatically adding incentives to dollars spent encourages additional spending rather than efficiency.

Options for New Resources



Setting Rates

Any decision regarding future energy resources will be reflected in the prices customers pay for electricity. If one alternative costs more than another, prices to customers will reflect that difference. If the resource selected is less expensive than the current average and everything else stays the same, then rates could go down.

The time of use and the amount of electricity used by WTU's different customer groups varies

a great deal. As a result, it costs more to provide electricity to some groups than to others. To a large extent, the PUCT assigns costs to the various customer groups based on the cost to provide service to that group. Assigning these costs is known as "cost allocation."

One of the most common allocations is based on the percent of peak load (the highest rate of electricity consumption over time) of the various customer groups. Industrial customers may use many more units of energy, but they may be able to lower the amount of electricity used during the peak hour. Consequently, some industrial customers may pay a lower price per kilowatt-hour than other customer groups.

Currently, residential customers make up the customer group with the highest cost per kilowatt-hour. These customers significantly contribute to the peak need for electricity because their energy use is driven by air conditioning, heating and hot water used during the peak in the afternoon.

Averaging

Averages are used extensively in setting utility rates. One house or business or factory may be 10 miles from the power plant and another one might be only one mile away. Both customers pay the same rate, yet the customer who lives 10 miles from the power plant may be more expensive to serve. Therefore, everyone in the same rate group pays the same rate based upon their average use.

The IRP Process

Last year the Texas Legislature passed a law requiring utilities to conduct integrated resource planning (IRP). Below is an overview of the IRP process that the Commission has implemented.

1. WTU identifies future need for additional capacity.
2. WTU announces the need and gathers information from the public. *(This is the step that WTU is now in.)*
3. WTU prepares and files a preliminary plan with the PUCT.
4. A hearing is held and the PUCT approves, denies or modifies the preliminary plan and WTU's process for inviting other suppliers to bid to supply the need.
5. WTU conducts the solicitation and selects contractors to provide the service or WTU decides to construct a plant.
6. WTU files a final IRP and asks the PUCT to approve purchase power contracts or to grant it permission to build a new power plant.
7. A hearing is held and the PUCT grants or denies the request.

The Increase Efficiency Option

It is feasible and often less expensive to meet some of the need for power by using the energy we already have more efficiently. Energy efficiency can mean making homes and businesses use less electricity with insulation, window shading, energy-efficient lighting, energy-efficient refrigerators, water heaters and air conditioners and using more solar energy in buildings. The purpose of energy efficiency is to decrease the amount of energy used without compromising the quality of service the customer receives.

These changes and investments take place on the customer's side of the meter where the demand is, and are, therefore, known as demand-side resources. The use of demand-side resources to meet new generation needs has boomed in the last 10 years as new technology in lighting, motor drives and controls has entered the market. These technologies are changing and improving.

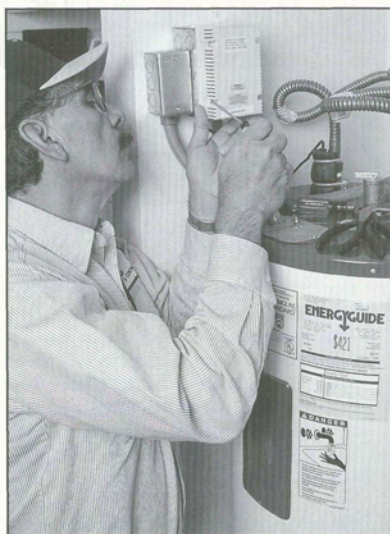
WTU offers customers incentives for using energy more efficiently. Some utility observers say efficiency programs raise rates. Others say that efficiency programs save energy, protect the environment and avoid new generating plants.

Peak Clipping

One option to meet the need is to focus on the peak usage times — those hours and days when total usage is at its highest level — and try to reduce demand at these times when electric costs are highest. Normally, during the weekdays of July and August during the hours from five to seven in the evening, power usage is more likely to be at its highest for the year. This is because air conditioners are working almost constantly to cool off homes, stores and dormitories. Also, customers come home from work and are cooking dinner and washing clothes.

As the population in the region increases, WTU's ability to meet this peak demand reliably will diminish. Eventually, the company will be unable to meet the need without more resources. Since peak usage is usually what causes utilities to need more power, reducing peak loads can delay the need. If demand can be reduced at this peak time, the need to acquire new resources may be delayed.

Some programs reduce peak load by charging higher rates during periods of high demand. Others reward customers for allowing the utility



Encouraging customers to purchase energy-efficient appliances and to install energy efficiency controls, such as timers, help control a utility's load growth while lowering the customer's bill.

to interrupt their service when necessary. Examples of peak clipping programs are control systems that set air conditioners, water heaters and refrigerators to run at different times so that all do not run at the same time. The customer seldom notices, but peak load of the overall system is reduced.

Programs must be carefully designed, because often residential and commercial users may have difficulty investing in equipment with high up-front costs, getting essential information, or taking advantage of programs aimed at peak reduction because it would mean changing their lifestyle.

Industrial customers are often in a better position to make use of these programs because they can either switch to backup generation that they own themselves or adjust their work processes to reduce demand. Some people say that focusing only on reducing peak usage misses a number of other cost-effective conservation measures. They believe that there are many other cost-effective options for using energy more efficiently. These energy efficiency options reduce utility costs, conserve fuel and are beneficial to the environment.

Energy Efficiency

Most customers can conserve electricity and still receive the same level of service. This is accomplished by making the building structure more efficient by installing insulation, caulking and weatherstripping, solar screens, roof overhangs and other measures that reduce cooling and heating needs. Another way to reduce electricity use is to install the most efficient air conditioners, water heaters, refrigerators, lighting fixtures and other equipment. Improving the structure reduces the size of cooling and heating

equipment. This reduces peak demand and lowers customer bills. The utility and its customers benefit by avoiding power plants and fuel costs. Emissions are reduced because less fuel is burned.

Some prefer peak reduction programs to energy efficiency programs because energy conservation reduces revenues and the loss of sales may raise rates. When the utility sells fewer units of electricity, the cost of the existing utility system must be spread over fewer units — so rates may go up.

Consumer groups and environmental groups assert that rates rarely increase because of lost sales due to energy conservation programs. If there is any increase, it is temporary. When a system is growing the saved kilowatt-hours are sold to someone else. Consumer groups and environmental groups prefer efficiency programs over peak reduction programs because they reduce customers' bills and save fuel. The fuel savings reduce customers' bills, save fuel and reduce air emissions.

Load Leveling

Some energy management programs are designed to even out the levels at which customers use electricity. Load leveling is also referred to as load shifting or valley filling.

An example of load leveling is a cold thermal storage system. The system chills water overnight and the water is used during the day to supply air conditioning. Storage systems shift the peak energy use of large buildings to off-peak hours.

Another type of program used to level loads on the power system is designed primarily to increase off-season sales. Sales are lower in the winter when the power plants built to keep air conditioners operating in summer produce less revenue. An increase in units of electricity sold without the need for new generating plants means the rate per unit can go down. These strategic load building programs, therefore, help support the sales of heat pumps, electric water heaters and other electric appliances mostly targeting the new home and apartment markets.

Most utility companies favor load leveling. Conservation programs have the company in the position of encouraging customers to use less of their product. Load leveling, on the other hand, results in more regular power plant operating hours and increases the kilowatt-hours sold from these plants. Consumer groups and environmental groups are less enthusiastic about load leveling. They reason that load leveling programs increase electricity use which means higher bills for customers even if the rate per kilowatt-hour is lower. Leveling loads also increases fuel use and emissions.

Load leveling can be accomplished by waiting several hours to use energy. Examples are washing clothes at night, starting an evening shift at an industrial plant or electric heating load.

Again, by lowering the peak, these programs make it possible to avoid the addition of generation. Another advantage to these programs is that customers can still use power for all the things they're accustomed to doing, but they just do so at different times. Some say that load leveling programs do not reduce fuel consumption and, therefore, do not help the environment. Others say that it allows the utility to use its resources more efficiently and that overall electricity rates remain the same or are reduced.

Equal Treatment

Fair treatment for different customer types has long been an issue with demand-side programs. Many of the programs require some action on the part of the consumer, such as an investment in more efficient equipment or an investment in a new more efficient appliance. The company, in turn, pays a portion of that investment as an incentive. Customer groups with low or fixed incomes often lack the funds to participate in the program even though it might be in their long-term economic benefit.

Another group that can easily be left out of these programs are tenants who comprise a substantial portion of WTU's residential customers. Investments, such as increased insulation that might make sense to the owner of a structure, may make little sense to a person who plans to spend a year or less living at that location.

One of the issues in this Deliberative Polling™ exercise is the extent to which customers are willing to support programs designed to achieve fair treatment of all customer groups. The utility could just meet statutory requirements or could go further.

Advocates state it is possible to design successful programs aimed at both renters and those on low or fixed incomes. They state it is a misconception that programs designed for low-income customers must necessarily be subsidized by other ratepayers. While it is true that some programs may cost more per unit, there are ways to design programs to minimize or eliminate the additional costs. For example, utilities can coordinate the delivery of low-income programs with other government-funded providers to eliminate additional administrative overhead and building costs (such as caulking and weatherstripping). Under current regulations, the utility company must design its programs so that all customers benefit equally. Some say that going further turns economic

Examples of WTU Demand Side Management Programs

Segment	Program	Focus
Residential	Good Cents New Home	High efficiency heating and cooling equipment; Promote energy efficient thermal qualities in new home construction.
	Good Cents Home Improvements	Replacing existing inefficient heating and cooling systems with high efficiency equipment. Increase the thermal efficiency of existing homes through insulation, storm windows, etc.
	Smart Money Financing	Financing for energy efficient technologies promoted through residential programs.
Commercial/ Industrial	Climate Control	Heating, ventilation, air conditioning, refrigeration equipment and technologies.
	Power Drive	Efficient electric motors and motor technology.

decisions into social programs. Others point out that, if low-income customers and tenants do not benefit from demand side management programs, at least to the extent of their financial contribution as ratepayers, they subsidize middle and upper income homeowners.

Pricing Options

Pricing can also influence when and how customers consume electricity. Higher prices during the summer often influence the amount of electricity used during that time of the year. These higher prices are necessary since it costs more to meet the demand in the summer when the peak is higher. There is also time-of-use pricing available for some classes of customers that reflect the changing cost of energy production through the hours of the day and the days of the week. These changes in price can also influence the on-peak usage by customers.

There are also other pricing options that can help reduce peak consumption and demand:

- Interruptible and curtailable rates are offered to large industrial customers where the customers accept the possibility of interruption or curtailment in return for a lower rate.

- Real Time Pricing will be implemented in the near future for different customers classes that have the ability to adjust operations on a daily or hourly basis to move load from the higher cost hours to the lower cost hours of energy production. This can lower their electrical bill and provide load leveling benefits to the company.

Historically, these programs have been aimed at large commercial and industrial classes. Some say that these special rates shift cost to the other classes of customers.

Glossary

Increase Efficiency and Conservation - Reducing energy use without changing lifestyle.

Peak Reduction - Using less energy during peak hours. Usually this term implies a change in lifestyle or business practices.

Thermal Storage - Changes electricity into another energy form such as chilled or heated water for use in heating or air conditioning at a later time.

Interruptible Rates - A rate structure where customers accept the possibility of interruption of service in return for lower rates.

Base Load - The level of utility load expected to be present most of the time. A baseload plant is designed for continuous operation and generates electricity around the clock.

Intermediate Load - The portion of utility load that varies between base and peak load. Intermediate plants typically operate 20 to 60 percent of the time.

Peak Load - A utility's maximum load during hours when rate of usage is the highest. Peaking plants typically operate less than 20 percent of the time.



Installing insulation during construction can reduce energy usage.

The Buy Power and Transport Option

A third group of options is to buy from outside the service territory and transport the electricity into WTU's system.

There is currently an excess of power available in many parts of the country, including Texas. This option has the advantage of allowing WTU to purchase only what it needs to meet growth on a year to year basis, and would not result in a long-term investment which could become uneconomic in the future. WTU has sufficient transmission capacity to transport the power in from other areas without building additional transmission lines.

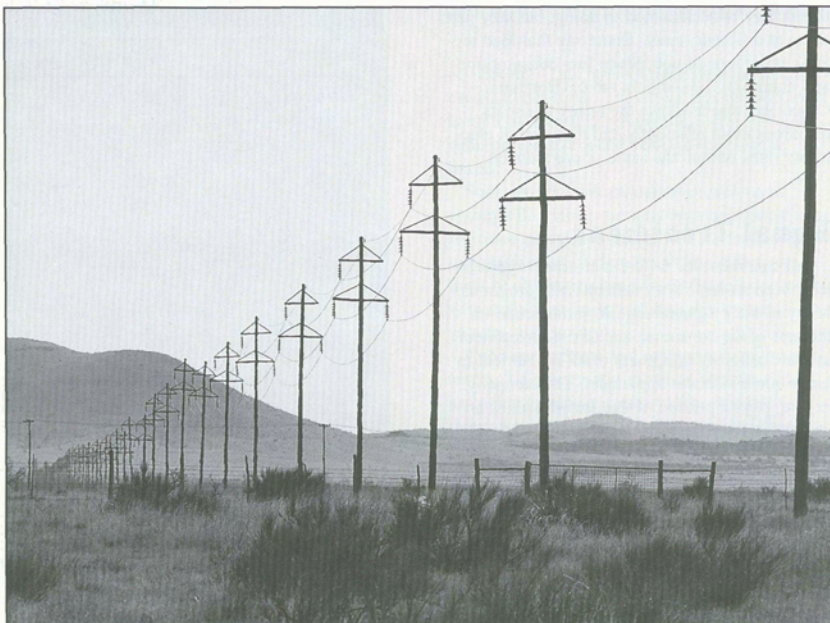
Power Purchases From Outside Sources

The electric utility industry is changing rapidly. More suppliers are entering the electricity market. These suppliers include power marketers, independent power producers (IPPs), other utilities, and cogenerators. As these suppliers enter the market, there are opportunities to take advantage of the excess power at low prices.

Many experts believe that the electricity market will continue to develop as more utilities rely upon this option for meeting the growth needs of their customers. Some suppliers have already begun to construct new generation facilities with the expectation that they will be able to sell their power into this market in the future.

Purchasing this excess power through the Buy Power and Transport option has the advantage of allowing WTU to purchase only what it needs to meet growth on a year-to-year basis, without having to invest in more long-term facilities which would be expected to be used and paid for over a 30-year period.

A disadvantage of purchased power is the long-term relationship between the utility and its customer is changed. Another company is generating the power and the utility is delivering it. While other utilities have purchased power on a long-term basis for many years, and the change in relationship has not been a problem to date, less than 10 percent of the power in Texas comes from non-utility sources. Another disadvantage is the difference in our ability to predict prices between long-term and short-term power purchases. Short-term purchases are favorably priced right now because there is a surplus. However, over the long term, we have less certainty on the price and supply of this power.



Transmission lines carry electricity from the generating stations to the local distribution system.

Power Suppliers

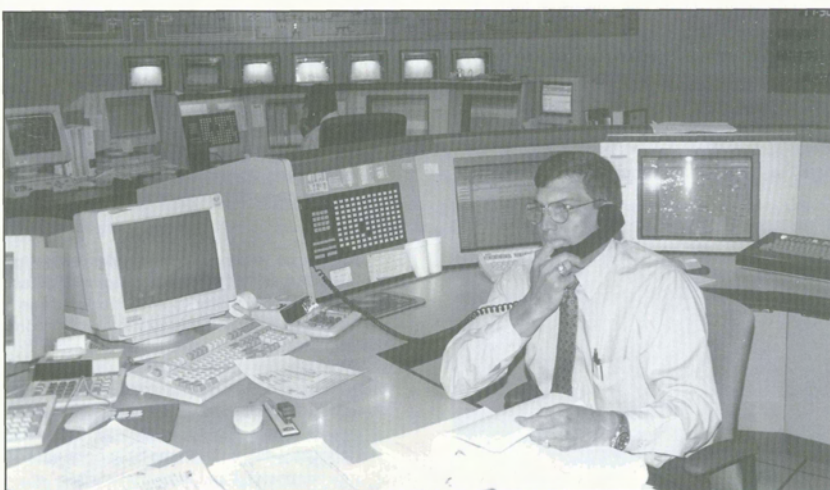
In addition to other utilities, WTU can turn to several non-utility sources to purchase bulk electricity.

Power Marketers

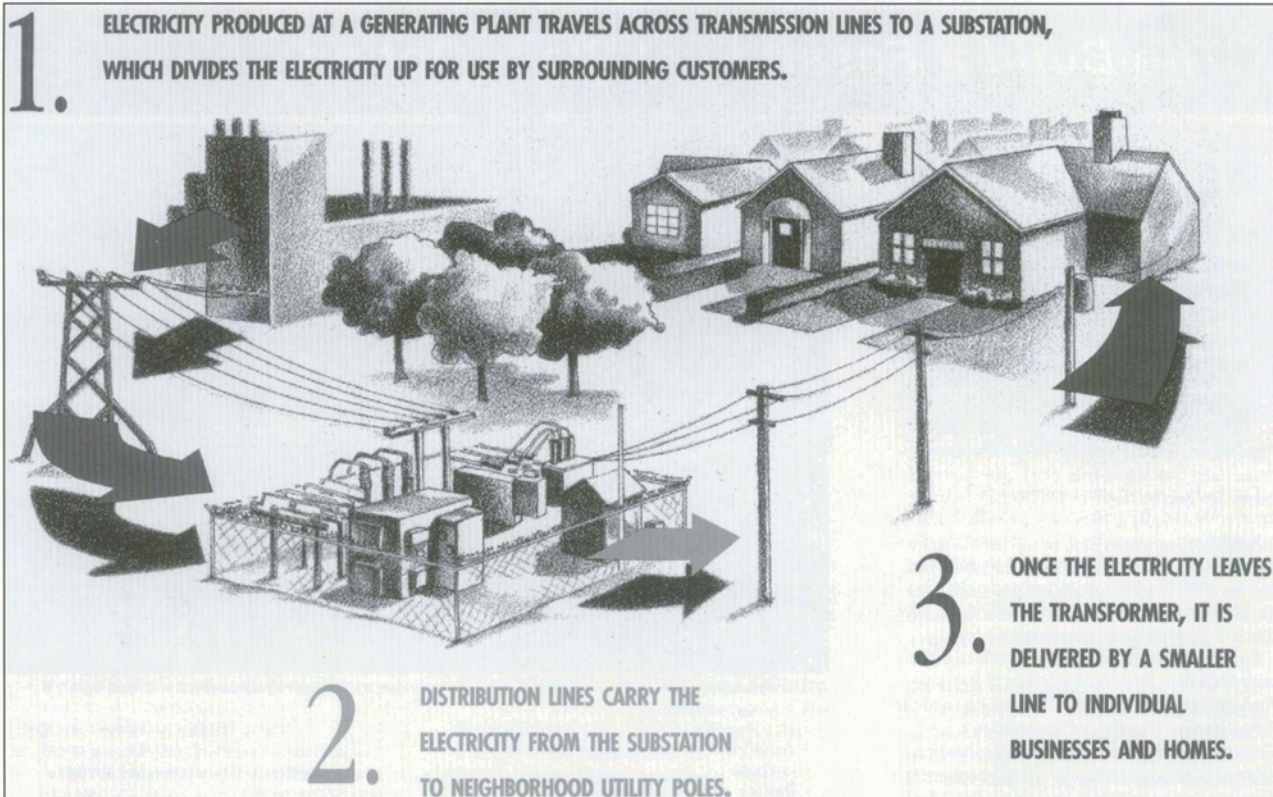
A new type of merchant has emerged in the electricity business. Just like the gas marketers that currently sell the bulk of the natural gas traded at the wholesale level, power marketers buy and sell bulk electricity, generally without owning generating facilities themselves. For example, power marketers can collect small

amounts of surplus electricity from a number of sources and put them together into a package to sell to others. A few years ago, there were only a handful of registered power marketers in the U.S. Today, there are over 170 registered power marketers.

Power marketers usually have a wide portfolio of electricity contracts available to sell. They are often able to sell a package that exactly meets the purchaser's requirements at a competitive price. The disadvantage is the weakened ability to predict the price and availability of electricity in the long term.



A utility's dispatch center is responsible for transferring the power from the generating stations through the transmission and distribution system.



Independent Power Producers (IPPs)

IPPs are businesses which own generation facilities for the production and sale of electricity to wholesale customers, such as electric utilities. There are several hundred large independent power producers in the U.S. today. Some are industrial plants, some are subsidiaries of construction companies and some are utility companies competing outside their service territory. CSW, the parent company of WTU, has a company that sells power in the independent market.

Cogeneration

Another source of electricity available to WTU is from cogeneration plants. These are plants that have a dual purpose: they produce steam and use it both to generate electricity and for steam in manufacturing processes. Usually these plants exist at major industrial sites, refineries or chemical plants.

Currently, there are no cogenerators in the WTU service territory. However, cogenerators do exist in other parts of Texas. Consequently, a cogeneration project may be a bidder under the Buy and Transport option.

The advantage of cogeneration is energy efficiency because more of the heat content of the steam is captured for energy use. Another advantage is economic, since cogeneration can help make the industrial steam user more competitive.

As long as natural gas prices are low, cogeneration may cost more than peaking natural gas generation.

Risk Management

Another important factor which WTU must consider when adding new resources is risk. The goal is to develop a portfolio, or mixture, of different types of resources such that the value of the mixture does not change unacceptably when external factors change. Two of these external factors are fuel price and the amount of power needed.

This situation can be illustrated through a simple example. Suppose you are in the delivery business for a perishable product like produce. The challenge is to match the number of trucks to customer demand. Demand is higher in some weeks from others over the course of a year and is growing steadily from year to year. Trucks that sit idle cost you in purchase price, licenses, and insurance even if you can control the variable expenses such as gasoline and drivers. Customers whose needs are not met, even for a few days a year go elsewhere. You can buy trucks, lease trucks for a year, rent trucks for a week and, at times in an emergency, borrow trucks for a day. The challenge is to meet customer needs without exception and to have as little idle capacity as possible.

WTU currently owns the resources necessary to meet the current demand for electricity. However, due to increasing competition in the electricity business, we cannot be certain how much electricity we will need to meet future demand. WTU has historically been the power generator for West Texas. WTU supplies a number of cities and electric cooperatives at wholesale and those cities and cooperatives resell to their customers. The wholesale market in Texas is now very competitive and the cities

and cooperatives will have options to buy electricity from other suppliers when their contracts run out. Should WTU buy excess power from someone else on a short-term basis and thus maintain flexibility? Should WTU commit to additional long-term resources on behalf of our customers, even those who may not continue to buy from WTU in the future? We want your opinion.

Transmission Capacity & Delivery

A combination of transmission and distribution lines carry energy to the consumer. First, transmission lines transport electricity long distances and usually carry power to a substation. Then, the distribution system takes over and carries the electricity from the substation to the customer. All power plants require transmission lines to deliver the electricity out of the plant and onto the transmission network.

WTU currently has sufficient transmission capacity to transport electricity into the service territory without constructing new transmission lines. Current estimates indicate that WTU could potentially transport enough electricity into the service area to meet their growing demand for electricity. There are, however, specific areas within the WTU service territory which will require additional transmission capacity within the next ten to fifteen years to maintain system reliability.

The Build a Fossil Fuel Plant Option

As West Texas outgrows its existing stock of generating resources, one obvious response is to build a new power plant. The new plant could be built either by WTU or a third party. WTU and its customers the various kinds of power plants in determining whether this option should be pursued.

Currently, the two most common fuels for generation are coal and natural gas. Both are hydrocarbons formed by plants covered up millions of years ago and are referred to as "fossil fuels." Both are finite, meaning some day we will run out, but there is great debate about how much coal and natural gas remains.

Fossil-fuel power plants often produce more emissions than other options. The U.S. Environmental Protection Agency (EPA) and the Texas Natural Resources Conservation Commission regulate the amount of emissions allowed. There are pollution-control equipment costs associated with building a new fossil fuel power plant.

Natural Gas

Natural gas is the most common fuel used to generate power in Texas, comprising approximately 65 percent of the total energy produced by WTU.

There are three basic types of conventional natural gas technology:



WTU has six power plants that use natural gas to generate electricity.



WTU's Oklaunion coal plant, located near Vernon provides approximately 35 percent of the company's power generation needs.

- Traditional steam generator-steam turbine
- Simple cycle gas turbines
- Combined cycle

Traditional steam generators linked with steam turbines have been the technology of choice for large natural gas plants, in the range of 200-400 megawatts. Their size and time necessary to start up makes them less useful for start and stop operation in meeting peak loads. Conversely, their efficiency in fuel justifies the higher equipment costs when the unit is run all the time.

Simple cycle units are the least expensive and use energy converted from natural gas only once.

Simple cycle units can be turned off and on easily and are therefore ideal for use during peak times of the day or peak periods of the year. Although they use more fuel to generate one unit of electricity, it is often economical to use a simple cycle if it is only needed during peak times.

Combined cycle units capture the exhaust gas from the simple cycle gas turbine and use it to make steam to power a second generator. The tradeoff is the cost of equipment versus the cost of fuel. Combined cycle units are more complex and, therefore, more expensive but more efficient. Some plants start out as peaking plants with a simple cycle and are later converted to combined cycle as the load in the area grows.

There is still considerable debate on the amount of natural gas available. As recent as 1978, natural gas was outlawed as a fuel for new generation. Some people point out that natural gas has many manufacturing uses and say that it should be conserved. Others say we currently have a surplus and can find more gas with new technology.

The federal law restricting the use of natural gas as a boiler fuel was amended in 1987 to allow utilities and others to construct new power plants using natural gas and repower existing power plants fueled by natural gas. This factor, along with a surplus of inexpensive natural gas, has made gas generation the technology of choice for most recently constructed generation plants.

A natural gas plant costs less than a coal plant and can be built in a much shorter time period. However, natural gas prices have historically fluctuated more than coal. Improved financial tools allow natural gas customers to better control these price fluctuations than they did in the past. Below are some of the advantages and disadvantages of natural gas generation.

Advantages of Natural Gas Generation

- Least expensive to construct
- Flexibility in size
- Short construction time
- Burns cleaner than other fossil fuels
- Less pollution-control equipment required
- Fuel is piped directly to the plant; no fuel handling required

Disadvantages of Natural Gas Generation

- Availability and price of natural gas can shift suddenly
- Some air emissions
- Some types of natural gas plants may require water for cooling
- Depending on the location, transmission lines may be needed
- Visual aesthetics

Coal

Using coal to generate electricity is a tested and proven technology, and coal is relatively inexpensive. Coal comprises approximately 35 percent of WTU's generation. Coal generation can use low-sulfur western coal transported by rail or can be a Texas lignite plant built near the mine. The newest coal plants use advanced fluidized bed combustion to reduce environmental concerns. Consequently, coal is an option to consider. Below are some of the advantages and disadvantages of coal:

Advantages of Coal Generation

- Historically, coal has been a cheaper fuel than natural gas
- More stable fuel price
- Plants can be designed to convert coal into natural gas, if necessary
- Texas is rich in lignite, a low-grade form of coal

Disadvantages of Coal Generation

- Is the main generating fuel that produces SO₂, which is linked to acid rain
- Is the primary source of CO₂, which many link to global warming
- Takes two to three times as long to construct as a natural gas plant
- More expensive capital costs
- Greater financial risk that construction will cost more than anticipated
- Needs more fuel handling and pollution control equipment, including coal and coal-ash handling facilities
- Needs water for cooling and disposal facilities for ash

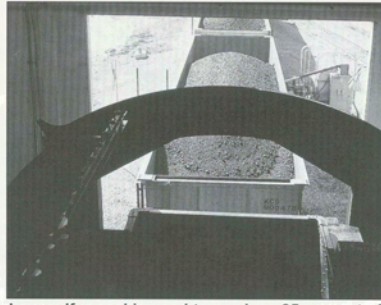
Distributed Generation

The use of small generating units at or near the site of electric utility customers is called distributed generation. Distributed generation can include on-site renewable energy, such as solar panels. It can also include combustion-driven generation, such as diesel units.

Today, distributed generation is already used at places that need to have backup power: hospitals, airports, water treatment plants and computer centers. These facilities typically have a generator available to use in case of a power failure. Customers who are billed, in part, based on their peak usage may use distributed generation to ensure that they do not set a new peak and increase their monthly bill.

Some say that small units are less efficient. Others say that distributed generation results in less investment in transmission and distribution lines. It also reduces the amount of "line losses" that increase as the distance of the transmission and distribution lines increase. In addition, rapidly advancing technology is making many small distributed generation sources more efficient than ever. Costs vary widely depending on size, location and application of the resource.

Other Considerations



Low-sulfur coal is used to produce 35 percent of WTU's electricity.

Nuclear Generation

Like coal, nuclear generation flourished following the 1972 Arab Oil Embargo when natural gas prices skyrocketed and utilities were incorporating fuel diversity into their strategic plans. In 1978, the Industrial Fuel Use Act prohibited using natural gas as a boiler fuel. But nuclear generation suffered a huge setback following the Three Mile Island nuclear incident in Pennsylvania. At this time, nuclear generation is not an option being considered.

Generation Storage

One of the problems with electricity is that it is difficult to store. Recent developments in energy storage include these two types:

- pumped hydro storage
- compressed air storage

In pumped hydro storage, water is pumped from a lower lake or containment area such as a salt dome, to a higher lake or containment area at an off-peak time when there is extra generating capacity. When the peak need arrives, the water is released down a shaft where it turns a turbine on the way to the lower containment area and produces electricity.

The same principle is true of compressed air storage. With the shortage of water in West

Texas, compressed air storage may be more likely. At an off-peak time, when there is extra capacity and fuel prices are low, electricity is used to compress air which is stored in an underground cavern. When the peak arrives, the compressed air is released and drives a turbine to generate electricity on peak. Storage might be considered a part of the long-term plan for WTU.

New Developments

Several new developments in generation look promising but are still in an experimental phase. Most prominent in this category are fuel cells, which generate electricity with a chemical reaction similar to a battery and can use natural gas as the fuel. They are still more expensive than either conventional generation or wind power but their design and economics are rapidly improving. Their advantages are small size, efficient fuel use, low emissions and quiet operation. Most see their first applications as generation sources for remote individual buildings and places that require high power quality. Using fuel cells to meet part of WTU's 10-year resource need would be feasible only as part of a research and development package.

Existing Resources

Traditionally, integrated resource plans have focused on the development of an appropriate mixture of new resources to meet load growth. However, another important factor to consider is how to best utilize the existing resources. In particular, consideration should be given to the potential for efficiency and productivity improvements, repowering and refurbishment of power plants and extending the life of power plants. The question of existing unit retirement is very complex and should be analyzed based upon the value to the system as compared to other available alternatives.

WTU Power Plants

Plant Sites ⁽¹⁾	Fuel Type	Years Units Were Constructed	Estimated Retirement Date	MW Capacity	Annual Output (MWHs)	% of Time Used
Lake Pauline	Gas	1928 - 1951	2009 - 2010	40	728	2%
Fort Phantom	Gas	1974 - 1977	2020	337	1,672,955	57%
Rio Pecos	Gas	1927 - 1969	2001 - 2015	127	779,614	70%
Oklaunion	Coal	1986	2021	363	2,312,290	73%
Paint Creek	Gas	1953 - 1971	2010 - 2015	218	507,532	27%
Oak Creek	Gas	1962	2008	75	395,791	60%
San Angelo	Gas	1965 - 1966	2010	110	857,193	89%
Various Gas Turbines	Gas	1916 - 1952	2009	34	91	<1%
Renewables	Wind/Solar	1995	2025	6	13,000	25%

⁽¹⁾ Each plant site may contain more than one generating unit.

The Build Renewable Energy Plants Option

The previous two options focused on electricity generated from fossil fuels, such as coal and natural gas, sources that will eventually diminish. As an alternative, we could consider those sources of energy that are not limited. The most common types are wind power and solar energy.

Wind Generation

Wind turbines are beginning to come of age. In the last five years, wind power has made major technological and economic strides. It compares favorably to natural gas because we can't be sure how much natural gas will cost in the future. West Texas has some of the more favorable sites for wind power in Texas.

Advantages of Wind Power

- Least expensive form of renewable generation
- Flexibility in size
- Serves as a hedge against sudden shifts in the price of fossil fuels or changes in environmental regulation
- Can help provide additional increments of electricity during peak electricity use.
- May be competitive with natural gas generation over the 30-year life cycle of the wind generation equipment
- Has no emissions

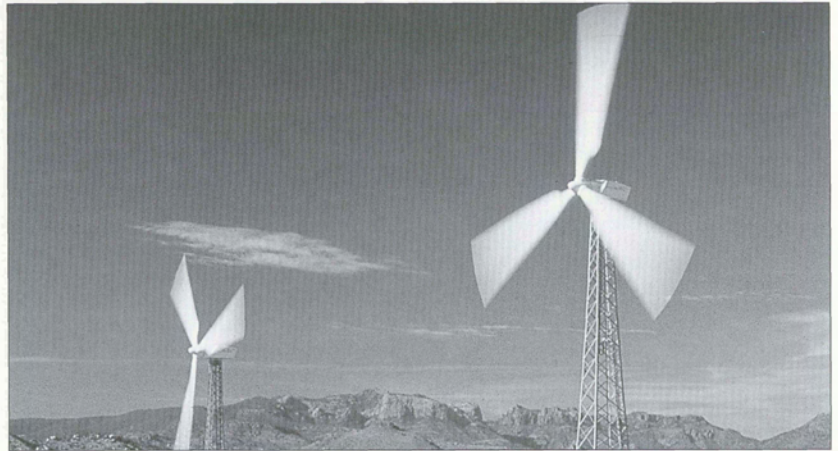
Disadvantages of Wind Power

- The wind may not be blowing when it is needed
- Possibility of birds flying into the blades
- Visual aesthetics
- Initial cost per kilowatt-hour is higher than fossil fuel
- Depending on the location, new transmission lines may be needed.

Availability of Wind Power Within WTU's Service Territory*

	1994	1995
Roxa (Fort Davis)	22%	23%
Trent (Abilene)	26%	28%
South Mesa (McCamey)	42%	41%

* Figures reflect the results of wind measurements taken at these sites during the last two years. No corrections made for losses due to turbulence, line losses, blade soiling, etc.



In the Davis Mountains of far West Texas, Central and South West Corporation is conducting an extensive research project to measure the potential of solar and wind power as alternative energy resources. In addition to 12 500-kilowatt wind turbines, the project includes the installation of wind measurement stations across the corporation's 152,000-square-mile service territory to locate the best wind sources.

Solar Power

Solar power is the use of the sun to meet energy needs. It can be used in many different ways to meet customer needs for heating and lighting buildings; for hot water in homes, businesses, and industry; and to make electricity, either for sale to customers or directly for use by customers.

Passive Solar and Daylighting

Passive solar building design means using a building's orientation to the sun, as well as strategic placement of windows, overhangs, and other building features to minimize heating and cooling loads. Daylighting means designing a building in order to bring the sun's light into building spaces. This could include new reflective technologies. In both cases, electric loads for heating, cooling and lighting can be minimized or avoided.

Solar Thermal

The sun's heat can be collected to heat water or air for use in homes, businesses or industry. Where these solar collector technologies are used, they may replace or reduce the size of furnaces or hot water heaters or they may be used to preheat water for commercial or industrial uses. Solar hot water heating was popular during the 1970s and early 1980s when natural gas prices were high. Adding solar collectors for heating buildings and for hot water will reduce the amount of electric energy required.

Solar heat can be collected on a larger scale and used to power electric generators. In certain California desert areas, large fields of reflective troughs aimed at the sun heat oil within a network of pipes. The heated oil, in turn, transfers the "solar" heat to steam generators.

These generators also use some natural gas to make electricity when the sun is not shining. Another large solar power plant uses large fields

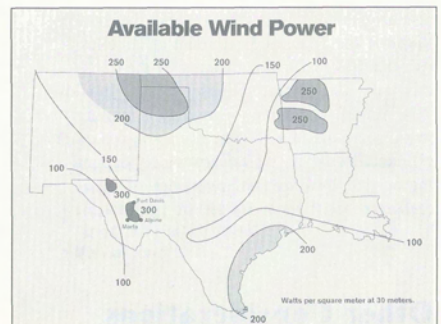
of mirrors focused on a central power tower to heat molten salt to temperatures which can create steam to drive an electric generator.

A different, smaller solar power plant is now being developed which involves mirrors focused on a special engine that uses the solar heat together with natural gas to produce electricity. These solar heat electric generation technologies are in various stages of development. Some are producing power for utilities today, while others need more time and testing, as well as further cost reductions before they will be ready for use by utilities.

Solar Electricity/Photovoltaics

Several very different technologies have the ability to transform solar power into electricity.

Solar electric power can be created by semiconductor materials in solar panels which change sunlight into electric current. Solar panels have been used in the space program from its beginning. Photovoltaic materials have many uses in consumer products, such as calculators, lights and battery chargers. Other currently cost-effective uses for photovoltaic electric power include water pumping, electric power for remote homes and for any small electric load that is remote from, or hard-to-serve, by conventional electric power lines.



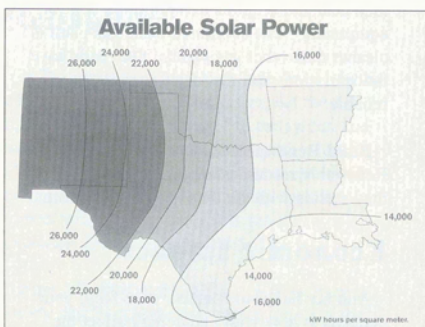


WTU's parent company, Central and South West Corporation, is exploring different ways to convert solar energy directly into electricity. Photovoltaic systems, such as the one at a Fort Davis drug store, produce approximately 2 kilowatts of electricity — enough to power the average West Texas home when the system is operating.

Solar panels on rooftops can help meet the building's peak electric load, which typically occurs when air conditioning needs are the highest. Solar panels can also add power to overloaded distribution lines and delay the need for a transmission investment.

Advantages of Solar Power

- Pollution-free
- Flexibility in size
- Uses the sun's energy as a source of heat to generate electricity
- Avoids the costs and risks of fossil fuels
- Cost-effective for specialized uses
- Supplies power during peak hours



Disadvantages of Solar Power

- Capital costs are higher
- Because weather conditions may limit the amount of sunshine, the utility must combine solar energy with other types of power
- Visual aesthetics

Green Pricing

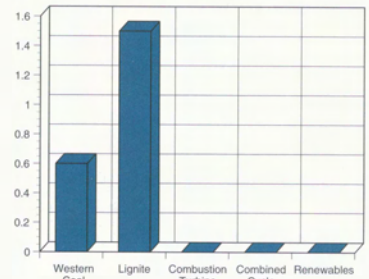
"Green pricing" has been around for a while and is now receiving renewed attention as a way to keep investments in renewable resources while facing the realities of competition in the marketplace. Several national polls report that some customers are willing to pay more for improved environmental performance. Green pricing is an attempt to allow those customers to exercise that choice.

Many electric utilities are examining the feasibility of designing green pricing programs that will be introduced in the future. The most important variable is a mixture of customer viewpoints. These customers pay the bills. For this reason, accurate marketing of green pricing is essential.

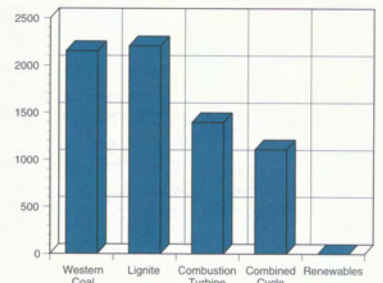
Will some customers pay more? What do they expect in return? Do they need an individual benefit or is an overall societal benefit sufficient? This is one of the issues WTU hopes to learn more about in the Deliberative Poll™.

Emission Comparison of Power Plant Options

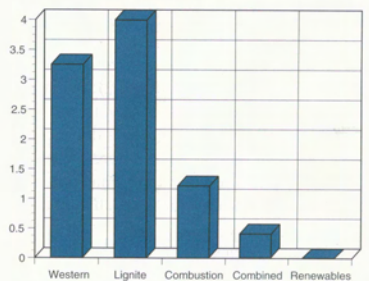
SO₂ Emissions
(Lbs./Mwh)



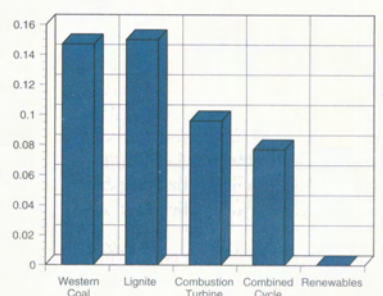
CO₂ Emissions
(Lbs./Mwh)



NO_x Emissions
(Lbs./Mwh)



Particulate Emissions
(Lbs./Mwh)



Planning for the Future:

This is Where We Need Your Judgment

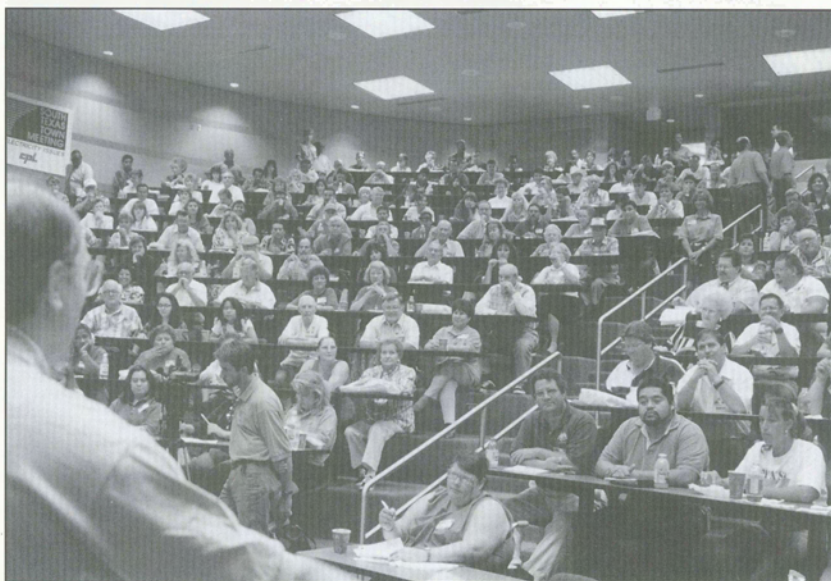
The 10-year plan for WTU's future will consider what WTU customers think is important.

It should also be a plan that can perform well under several different futures, since the future is uncertain.

You've now read about the options. Now we're asking you to tell us what mix of these options you would like. In reaching your decision, you must consider the advantages and disadvantages of all the options including:

- **Dollar Cost**
- **Environmental Cost**
- **Reliability**

You as a consumer make these types of tradeoff decisions every day. Take for example, the tradeoffs that must be considered when purchasing a new car. What type of car can you afford? What type of gas mileage will it get? How much insurance will you have to pay? How much maintenance will one car need as compared with a different model? Depending on your situation, the "right" solution will likely involve a tradeoff.



Central Power and Light Company (CPL), another Central and South West Subsidiary, was the first company to ever use a Deliberative Poll™. CPL held its Deliberative Poll™ in June.

chased from the outside. Prices are currently low, but will they be low in five or 10 years? You can get a long-term purchased-power contract, but the price tends to be higher for short-term contracts because the seller takes the risk. WTU's current generation mix is heavily weighted toward long-term contracts; so, some argue cheaper short-term contracts are the approach to pursue.

the environment. Some customers prefer resources which do less harm to the environment. Other customers feel it is not worth the additional cost for the utility to go beyond state and federal compliance. Environmental advocates argue we are under-estimating costs associated with conventional power plants, such as global warming and health effects.

Dollar Cost of Resources

Just as cost is important in purchases you make every day, cost is a driving force in selecting energy resources. Obviously, the dollar cost is a major influence on a utility's decision making. Different options have different costs associated with them. Let's examine some of the costs you should consider for each option.

Increase Efficiency: With some programs, there is an initial cost associated with installing energy-efficiency equipment. Energy efficiency programs are designed so that the utility pays a portion of the costs of energy efficiency. The bulk of the investment is made by the customer. Programs designed to use less electricity mean the utility sells fewer units of electricity. If the utility has excess capacity, and costs are spread over fewer units, rates may increase. In a situation where the utility needs to add capacity, there is no increase in rates. The saved kilowatt-hours are sold to other customers. Customers who employ more efficient use of electricity are rewarded by lower bills.

Buy Power and Transport: There is a surplus of power in Texas. Current prices are low, but the market is beginning to heat up. It is difficult to predict market prices for power pur-

Build a Fossil Fuel Plant: Gas plants are the least expensive plant to construct. Coal plants are more expensive to construct than gas, but coal is a cheaper fuel than natural gas. Both natural gas and coal plants are well-known reliable technologies, but both have indirect costs.

Fuel prices are difficult to predict. In the past, experts said fuel prices were as low as they could go. But they were wrong. Fuel prices went lower.

When fuel supplies are limited, prices rise. Prices also can rise because of increased production costs, higher transportation costs or rising taxes. One of the largest areas of uncertainty concerns when to lock in a fuel price.

Build Renewable Energy Plants: The construction and other capital costs for renewable projects are higher than for other fossil fuel options. However, there are no fuel costs associated with renewables. Renewable energy can be cost-effective for specialized uses today. Prices for renewable energy can help offset expensive peak generation.

Environmental Costs

Environment

There are state and federal requirements that a utility must comply with in order to protect

Increase Efficiency: Efficiency programs, which reduce electricity use are good for the environment because they reduce the need for fossil fuel used to produce electricity. Programs that shift electric loads to off-peak times have no effect on the amount of fossil fuel burned.

Buy Power and Transport: WTU currently has sufficient transmission capacity if the decision is to purchase power from the outside. Cogeneration and power marketing have similar environmental concerns as the Build a Fossil Fuel Plant option because they have emissions from a new or existing plant.

Build a Fossil Fuel Plant: Gas burns cleaner than coal and requires less pollution control equipment. The combined cycle gas plant is cleaner than other gas plants. Fuel cells have low emissions, but the technology is not yet reliable.

Build Renewable Resources Plants: Renewable resources are emission free. Birds may collide with the blades of a wind turbine.

Economic Impact

Another factor to consider in making your decision is how your choice will affect the economy and any potential for creating jobs.

Cost Comparison of Resource Options

Technology Description	Typical Size (MW)	% Available During Peak	Estimated % of Times Unit Will Operate	What It Costs to Build (\$/KW)	What It Costs To Build and Operate (Cents/KWH)	Time To Construct (Months)
Natural Gas Fueled Units:						
Simple Cycle Combustion Turbine	26 to 64	95%-100%	20%	\$270 to \$709	5.92 - 10.32	30 to 32
Combined Cycle Combustion and Steam Turbine	36 to 232	95%-100%	80%	\$452 to \$995	2.80 - 4.11	30 to 36
Solid Fueled Units						
Plant Burning Western Coal	335 to 740	92% - 100%	80%	\$1,037 - \$1,304	4.01 - 4.90	54 to 60
Plant Burning Lignite	333 to 740	92% - 100%	80%	\$1,070 - \$1,424	4.84 - 5.98	54 to 60
Renewable Technologies						
Wind Generation	50	<10%	25%- 40%	\$900 - \$1,200	4.2 - 10.0 ⁽¹⁾	12 to 18
Wind Generation w/Backup	50	95% - 100%	25% - 40%	\$1,116 - \$1,416	5.4 - 11.9 ⁽¹⁾	30 to 36
Solar - Photovoltaic Fixed Flat Plate	1 to 50	40% - 50%	25% - 40%	\$2,371 - \$3,295	26.93 - 33.38	24 to 36
Solar - Parabolic Trough/Gas Hybrid	80	NA	40%	\$2,882 - \$3,405	33.36 - 39.73	24
Increase Efficiency Note: This includes options such as residential appliances, heating and cooling equipment, commercial lighting, heating and cooling, and industrial motor replacement, variable speed motors	1 to 10	100%	100%	\$250 - \$1,200	0 - 7	12 to 60
Buy and Transport	variable	100%	N/A	\$40 - \$80/yr ⁽²⁾	2.5 - 5.0	—

Note: These are cost estimates. Actual costs of each alternative will be determined during WTU's bidding process.

(1) These costs include an investment tax credit of 1.2 cents per Kwh which is due to expire in July 1999.

(2) This is an annual cost — the other numbers in this column are one-time investment costs.

How important is the economic development factor in selecting WTU's resource mix?

Increase Efficiency: Conservation programs create jobs and typically direct money to local manufacturers and installers of high-efficiency equipment and insulation.

Buy Power and Transport: Power purchased from outside will have less of an economic impact.

Build a Fossil Fuel Plant: A plant built in the area creates construction jobs and some permanent jobs, and also increases the tax base.

Build Renewable Energy Plants: A renewable energy plant built in the area creates construction jobs and permanent jobs and also increases the tax base.

Reliability

WTU currently maintains a high level of reliability. We assume customers expect their lights to turn on when they flip the switch and that they would be upset with anything less. Currently, some large customers accept lower rates in return for lower reliability. Are customers willing to accept lower reliability in exchange for a lower electric bill?

Fuel Mix and Availability: Will there be a steady supply of coal and natural gas over the

next 10, 50 or 100 years? Will exploration prove that natural gas is plentiful? Do we want to use finite and depleting resources, such as oil and natural gas, when we have resources that can regenerate over time like wind power? You must wrestle with these questions when deciding on the proper mix of energy resources to provide reliable service. One way to reduce the fuel supply risk is to diversify your resources.

Wind and solar power are the best current examples of a resource not limited by supply. Depending on location and size of need, they are within striking distance of being cost competitive compared to some higher projections of future natural gas prices. Some customers argue they cannot afford the decision to use more renewable energy. Some customers say they could decide to leave the system to generate cheaper power on their own.

The fact that wind and solar power have an unlimited supply may not be enough reason to select it, but when coupled with environmental benefits, it may make sense to include it as part of the energy portfolio. Central and South West Corporation, WTU's parent company, has invested in wind machines with a total capacity of six megawatts in West Texas as part of an ongoing effort to gain experience with renewable generation.

Pricing: How important are stable, predictable prices to customers? Should special pricing programs be introduced that encourage investment in renewable resources?

Some Closing Thoughts

How much of the energy need should be met through increasing efficiency? How much should be met by building a power plant? Do you feel more renewable energy should be used? These are the questions that WTU needs your judgment on.

In making your decision, you need to consider the advantages and disadvantages of all the options. The right mix of resources to meet your concerns might contain one or more of these options. How much of which resources seems best to you?

About This Briefing Newspaper

These materials were assembled by WTU with the active assistance of an Advisory Group including the staff of the Public Utility Commission of Texas, the Environmental Defense Fund, Texas Ratepayer's Organization to Save Energy, Public Citizen, Consumers Union, Enron, and Texas Legal Service.

The materials represent a general consensus of the group although each participant may not agree with all sections. The goal was to present balanced information and a diverse range of viewpoints and opinions. WTU has the final responsibility for the materials.

An Overview of the Deliberative Poll™

Professor James Fishkin, chairman of the Government Department at the University of Texas at Austin, developed the concept of Deliberative Polling™ as an approach for determining informed public opinion. The approach received international attention during the National Issues Convention held in Austin in January 1996.

Deliberative Polling™ differs significantly from ordinary polls and surveys. As indicated in Professor Fishkin's book, *The Voice of the People, Public Opinion & Democracy*, ordinary polls and surveys examine what the public is thinking about different issues at a particular time. A deliberative poll attempts to determine

what the public would think about different issues if given an opportunity to learn more about all sides of the issues.

First, a poll is taken that determines public opinions about the issues under consideration. Then a scientifically random sample of the population is identified. That group meets in one location where they are immersed in the issues through carefully balanced briefing materials, intensive discussions in small groups and through the opportunity to listen to and question experts with dif-



Professor James Fishkin

fering views.

At the end of the process, the group is again surveyed to determine whether their views on the various issues changed and to what extent. The resulting survey offers a representation of the considered judgments of the entire population from which the sample was drawn.

This process is vital to WTU in that it provides important information about what values customers place on certain approaches or combinations of approaches for meeting future energy needs. The final plan that WTU submits to the PUCT could involve investments of \$100 million.

Schedule of Events

Friday, August 9

9:00 a.m. - 11:00 a.m.	Registration at Hotels
11:00 a.m.	Board Buses for Hardin-Simmons University
11:30 a.m. - 1:00 p.m.	Lunch at Hardin-Simmons Campus Center
1:00 p.m. - 1:45 p.m.	Welcome and Orientation
1:45 p.m. - 2:45 p.m.	Small Group Session #1 - Efficiency Options
2:45 p.m. - 3:45 p.m.	Small Group Session #2 Buy and Transport Options
3:45 p.m. - 4:00 p.m.	Break
4:00 p.m. - 5:30 p.m.	Large Group Session #1 - Efficiency Options and Buy and Transport Options
5:30 p.m.	Board Buses to Hotels
6:45 p.m.	Buses leave for Civic Center
7:30 p.m.	Dinner/Party - Abilene Civic Center
	Entertainment - Patsy Torres

Saturday, August 10

6:30 a.m. - 7:30 a.m.	Breakfast at Hotels
7:30 a.m.	Board Bus to Hardin-Simmons University
8:00 a.m. - 9:00 a.m.	Small Group Session #3 - Build Options (Fossil)
9:00 a.m. - 10:00 a.m.	Small Group Session #4 - Build Options (Renewable)
10:00 a.m. - 10:15 a.m.	Break
10:15 p.m. - 11:45 p.m.	Large Group Session #2 - Build Options
11:45 a.m. - 12:15 p.m.	Small Groups prepare questions for PUCT representatives
12:15 p.m. - 1:15 p.m.	Lunch at Hardin-Simmons Campus Center
1:15 p.m. - 2:45 p.m.	Large Group Session #3 - Discussion with PUCT representatives
2:45 p.m. - 3:00 p.m.	Break
3:00 p.m. - 4:00 p.m.	Small Group Session #5 Bringing It All Together
4:00 p.m. - 5:30 p.m.	Large Group Session #4 - Stakeholder Panel
5:30 p.m. - 6:00 p.m.	Closing and Thanks
6:00 p.m. - 6:30 p.m.	Complete questionnaire and board buses back to hotels
7:00 p.m.	Dinner provided at hotel restaurant for participants

Role of the Moderators And Facilitators

Each small group has a trained professional moderator. Their job is to help you walk through the costs and consequences of the various options, keep the discussion on track and to make sure all participants have a chance to ask their questions. Moderators are recruited from outside WTU in order to maintain neutrality during the process. In addition, each group has a WTU volunteer facilitator. Their job is to solve problems.

Ron Lehr, a former chairman of the Colorado Public Utility Commission, will moderate the large group sessions.

A Special Thanks To:

Hardin-Simmons University

KTXS News 12

University of Texas at Austin

Deliberative Polling™ Advisory Group

Public Utility Commission of Texas

Speakers and Panel Members